Supplementary Evidence Appendices

1. Claim-to-evidence map

Table A1: Claim-to-evidence map.

Claim	Status	Evidence	Boundary
Admissibility gate	Method claim	Gate definition and manuscript method section	Establishes an auditable screening rule; physical validity still depends on each relation's stated preconditions.
Relation-record executable checks	Workflow claim	relation records and validators	Some records remain protocol evidence pending matched SUT evidence.
Baseline admissibility contrast	Observed	Expert-MR, generic-MR, LLM-MR baselines and claim ledger	Quantifies an admissibility gap without ranking methods as competitors.
Three-arm complementarity / gate value	Observed	PointMLP three-arm report and detection-vs-accuracy metric ledger	admitted relation set 13/20 vs accuracy monitor 6/20 (2×2: 9/2/4/5, Wilson CIs); gate-admitted template 0% vs gate-rejected templates 100% baseline false positive; a measured gate value and complementarity, never a superiority ranking; consolidated across three converged SUTs (cylinder MGN 5/10 vs 3/10, airfoil 1 vs 2 disjoint, airfoil gate-rejected templates 4/4 firing on the correct SUT; C51).
Minimum-MR-SubSet external-scope audit	Observed secondary audit	External-scope audit JSON	external witness evidence from 70 real rows and three SciML/PDE true-fault-class rows with PASS_WITNESS status; does not add new primary SUT executions to this paper.
Minimum-MR-SubSet primary rerun	Observed external runtime rerun	Witness rerun report	PASS_WITNESS status, kstar = 6, four active true fault classes, max signature rank 2, collapse false; real held-out cylinder-flow MGN runtime, not a second architecture or dataset.
Minimum-MR-SubSet PINN primary reruns	Observed second-SUT/PDE reruns	PINN witness rerun reports	trained Burgers2D and Diffusion2D PINN witnesses, PASS_WITNESS status, kstar = 1 each, five active true fault classes each; one-seed witnesses, no cross-SUT rate.
Node-permutation sanity check	Observed pilot	Node-permutation metric ledger	One representation-contract path on one pilot case.
Conservation diagnostic	Diagnostic; absolute relation deferred	Conservation metric ledger and report	Reference-relative guard only; absolute conservation remains deferred.
Mirror-y OOD stress (PC6-mirror-y-ood-stress)	Observed bounded pilot	Mirror-y rate metric ledger and claim ledger	Failed on 10 of 10 recorded eval frames; not a reliability, accuracy, baseline, multi-SUT, exact-symmetry, or geometry-independent claim.

Claim	Status	Evidence	Boundary
Primary empirical scope upgrade	Observed	Primary-scope report and per-trajectory/checkpoint manifests	K=6 x 3 trajectories x 10 mirror-y OOD-stress grid fails 180/180; K=6 x 3 trajectories x 9 conservation-transition grid passes 162/162; K=6 x 3 exact-symmetric-mesh input grid fails 18/18; not a single-source-trajectory estimate and not a cross-SUT rate.
LLM role boundary	Process boundary	Method and ethics sections	LLMs organize candidates; the admissibility gate decides validity.
Rollout-accuracy diagnostic	Observed	Rollout-accuracy metric ledger	Same-SUT one-step median relative L2 0.0216; mirror-y is ∼34x larger.
Exact mirror-y symmetric mesh	Observed	Symmetric-mesh metric ledger	Exact relation admissible and fails (rel L2 1.10) on a synthetic OOD mesh; binary equivariance evidence.
PC10-seeded-fault-detection	Observed	Seeded-fault metric ledger	Detector stress test: MRs catch 5/10 injected mutants with class-specific response patterns.
Multicheckpoint replication	Observed	E1 aggregate and per-SUT manifests in `multicheckpoint/`	K=6 checkpoints of one MeshGraphNets family; within-family replication.
Operator-floor resolution	Observed	Operator-floor sweep report	Log–log slope 0.984, 95% CI $[0.975, 0.992]$ (interior 0.989), $R^2 = 0.9999$ over nine resolutions; reproduced on a second unstructured Delaunay topology (slope 0.983, floor ratio 1.06 median / 1.33 max), calibrating the admissibility predicate across two mesh topologies.
Shape-regular P1 operator-floor soundness	Supported theory	Phase B proof artifact and operator-floor reports	Local bound for P1 constant-per-cell divergence on shape-regular triangular meshes with C^2 divergence-free reference fields; supports numerical-decidability soundness for this operator class, not arbitrary degenerate meshes, non-P1 operators, or discontinuous fields.

Claim	Status	Evidence	Boundary
Fault-detection robustness	Observed	E3 robustness report; Phase-3 unified catalogue	30-trial Wilson CIs per MGN mutant plus a 60-entry unified fault catalogue (10 canonical MGN + 2 adversarial MGN + 24 closed-form PINN output-level probes + 24 closed-form FNO output-level probes), by-detector precision/recall with Wilson CIs, and Wilcoxon/Cliff effect-size tests; with realistic non-tailored faults the by-class structure emerges as a two-by-two partition (changes integral × breaks symmetry) on FNO and PINN, and a falsifiable coverage-completeness holds — every invariant-preserving fault is invisible to the admitted relation set with no falsifying case, blind even at 0.63 relative L2 on PINN; PINN/FNO probes are closed-form output-level probes.
S4/S5 variant primary workflow	Observed	Variant report and raw ledgers	Same-domain wider/deeper MGN variants: 2/2 node-permutation passes, 60/60 mirror OOD failures, 54/54 conservation-diagnostic passes, 6/6 exact-symmetric failures; not PhysicsNeMo/EchoWave or cross-dataset reliability evidence.
PointMLP cylinder primary workflow	Observed	PointMLP report, checkpoint, and raw ledgers	Newly trained non-MGN row-wise coordinate network: 9/9 node-permutation passes, 10/10 mirror OOD failures, 9/9 conservation-diagnostic passes, 3/3 exact-symmetric failures, median rollout rel L2 0.0298; not PhysicsNeMo/EchoWave or production CFD evidence.
PhysicsNeMo MGN Object-A smoke workflow	Observed smoke-subset production-framework execution	PhysicsNeMo smoke report, newly trained checkpoint, raw outputs, and metric ledgers	NVIDIA PhysicsNeMo `MeshGraphNet` executes on a first-record official DeepMind `cylinder_flow` smoke subset: node permutation passes (relative L2 0.0), mirror-y is OOD stress, and conservation is reference-relative diagnostic only; not full production-scale PhysicsNeMo, AeroGraphNet, or DoMINO evidence.
FNO primary workflow upgrade	Observed	FNO workflow report, per-SUT ledgers, and raw source/follow-up outputs	Six trained FNO-2D checkpoints over Burgers/heat: 24/24 translation passes, 24/24 conservation failures under a periodic discrete-conservation MR, and 6/6 Dirichlet-boundary translation rejections; full execution evidence beyond admissibility.

Claim	Status	Evidence	Boundary
Periodic-advection primary workflow	Observed	Periodic-advection workflow report, relation records, checkpoints, per-SUT ledgers, and raw source/follow-up outputs	Six NumPy-trained periodic-convolution SUTs on a synthetic 2D scalar periodic-advection task: 60/60 translation passes, 60/60 mass-conservation passes, and 6/6 fixed-velocity mirror rejections; full execution evidence on an independent synthetic PDE task, not production CFD or real-defect evidence.
RealPDEBench foil mirror-y preflight	Observed inconclusive preflight	RealPDEBench metadata screen, relation record, Arrow shard hash, and preflight metric report	Public real-flow zero-AoA NACA0025 foil field: coordinate mirror floor 0.0, u mirror relative L2 0.0600, sign-reversed-v relative L2 1.3753, combined uv relative L2 0.2292; typed verdict inconclusive because no independent PIV measurement-floor bound is available; not a SUT pass/fail, production-CFD validation, or real-defect claim.
External defect-witness corpus	Observed issue/PR/commit-linked witness corpus	Corpus summary, five relation records, external GitHub artifacts, and five witness reports	Five public external witnesses across four repositories/subsystems, including the DeepXDE PeriodicBC issue-linked witness: DeepXDE derivative periodicity, NeuralOperator `spectrum_2d` radial/power calculation, NeuralOperator Hermitian `irfft` boundary enforcement, PhiML custom-gradient axis order, and JAX-CFD flux-boundary inference; 5/5 typed pass; not representative defect sampling, trained-SUT correctness, production validation, or a defect rate.

2. Detailed design and boundary tables moved from the main text

The following tables were moved from the main manuscript during the JSS length repair. They preserve the review-facing evidence-unit, primary-result, and claim-boundary details while the main text retains only the compact narrative.

3. Cross-program breadth

Cross-program generalization. Reusing committed Minimum-MR-SubSet kill matrices, the coverage-geometry reading is checked across seven program types in three families: neural surrogates, classical numerical solvers,

Table A2: Closest-prior capability matrix.

Approach	What it supplies	What remains missing for the evaluation objective	What this paper adds
Scientific-software MR identification	Candidate relations from domain rules, source-code learning, LLM variables, and data-driven search	Candidate validity, numerical decidability, executable check format, and typed SciML verdicts	Treats these methods as candidate sources, not validity authorities
Reichert et al.	Physics-derived MRs and informal filtering in a learned hydrologic surrogate	Explicit admissibility predicate, operator-floor tolerance, reusable relation records, and typed verdicts	Formalizes the candidate-to-verdict path for mesh-based SciML surrogates
Eniser et al.	Relaxed MR oracles with numerical thresholds	Measurement-operator floor and relation-domain interpretation for deterministic SciML surrogates	Grounds tolerance in the scoring operator's own numerical floor
Duque-Torres / MetaTrimmer	Binary bug-versus-inapplicability separation with explicit MR constraints and automated constraint derivation	A typed inadmissibility verdict and operator-floor admissibility for physics-governed SciML	Replaces the binary skip/proceed pre-filter with a two-axis typed verdict tied to physical, geometric, boundary, and operator limits
Formal MR specification and constraint tooling	Machine-readable MR descriptions, constraint phases, and supporting systems	Physics-specific admissibility dimensions, measurement floors, and executable evidence ledgers for SciML	Uses specification ideas but binds each verdict to relation-domain and numerical-decidability evidence
Present study	Physics, expert, and LLM-assisted candidates plus artifact-backed execution	Bounded case-study evidence, not general SciML reliability or baseline superiority	Constructs relation records, runners, ledgers, and typed verdicts under a four-condition gate

Table A3: Evaluation design.

Item	Evaluation question	Evidence tier and unit	Artifacts	Claim boundary
Goal	Does the full workflow connect candidate, gate, executable check, verdict, and evidence?	Primary workflow trace; relation-level cell	Relation records, runners, claim ledger	Record/check construction only; no general reliability claim
RQ1	Which candidates are admissible, rejected, downgraded, or deferred?	Primary and supporting gates; candidate/MR	Gate records, operator-floor sweep, exclusions	Physical validity is relation- and SUT-specific
RQ2	Are admitted relations executable and auditable?	Primary execution cells; source/follow-up pair	Transformations, runner configs, metric files, logs	Execution completeness, not complete automation of MR discovery
RQ3	Do verdicts separate SUT inconsistency from domain and numerical limits?	Verdict ledger; MR-by-case cell	Two-axis verdicts, metric/floor ratios, exclusion logs	Per-relation interpretation; no cross-relation calibrated domain score
RQ4	What evidence is added beyond rollout accuracy and candidate-generation comparators?	Primary, supporting, secondary, and stress-test tiers	Rollout diagnostics, LLM/generic baselines, witness reruns, seeded-fault ledgers	Complementarity and bounded utility; no baseline superiority
Impl.	What blind spots follow from the admitted MR set?	Stress-test tier; mutant/MR cell	Seeded-fault catalogue and cross-SUT removal check	Bounded coverage interpretation, not real-world detection rate

Table A4: Independent evidence units: Nominal-N, effective-N, and inference permissions.

Evidence unit	**SUT/task and MR/operator**	**Independence source**	**Evidence role**	**Inference allowed**	**Inference forbidden**
Cylinder primary workflow	MGN cylinder-flow MRs	K=6 checkpoints, three trajectories, exact-symmetric inputs	Primary method execution	within-family typed verdict behavior	cross-SUT or population rates
Same-task architecture check	S4/S5, PointMLP, PhysicsNeMo cylinder	different impls. on one task	Implementation check	rules out single-codebase artifact	broad production-CFD or reliability claims
Changed-physics task	PhysicsNeMo airfoil gates	second CFD task, non-zero angle of attack	Gate discrimination	task-specific typed verdict change	SOTA or official-checkpoint claims
PINN/FNO PDE families	Burgers/heat operators	different representation families and calibratable floors	Cross-family predicate check	executable verdicts where floors are calibratable	neural-operator or real-defect rates
Periodic-advection primary workflow	six periodic-convolution SUTs	independent synthetic PDE/task, K=6 and 10 fields/SUT	Independent full-chain check	admit/reject gate-to-verdict execution	production CFD, real-defect, reliability, or broad generalization
RealPDEBench foil mirror-y preflight	public zero-AoA NACA0025 real PIV field	public RealPDEBench real-flow data, independent metadata and Arrow artifact	Production-adjacent preflight witness	metadata, coordinate map, and source/follow-up metric generation with inconclusive typed verdict	SUT pass/fail, real-defect detection, production CFD validation, or aerodynamic generalization
External defect-witness corpus	five component/utility witnesses	public issue/PR/commit artifacts from DeepXDE, NeuralOperator, PhiFlow/PhiML, and JAX-CFD	External real-defect/fix-linked witness corpus	five documented component-level contrasts represented as relation records, source/follow-up witnesses, metrics, and typed verdicts	representative defect sampling, trained-SUT accuracy, framework-wide correctness, production validation, or defect rate
Fault and external witnesses	seeded/probe catalogues plus sibling rows	stress catalogues and read-only external artifacts	Blind-spot stress and external audit	admitted-set coverage geometry	real-world defect rate or superiority

Table A5: Primary cylinder-flow result summary.

Check	**Observed result**	**Interpretation**	**Boundary**
Node permutation	Relative L2 = 0.0	Execution and inverse-mapping sanity check for a representation MR	Does not establish model accuracy or reliability
Asymmetric mirror-y	10/10 OOD-stress failures; median relative L2 0.737	The gate downgrades exact symmetry to stress evidence on this mesh	Not an exact symmetry verdict or geometry-independent fault claim
Continuity	Absolute conservation deferred; reference-relative ratio about 0.4–0.8%	Numerical floor blocks an absolute verdict while allowing a non-regression diagnostic	Not an absolute mass-conservation claim
Rollout contrast	One-step median relative L2 0.0216; mirror-y violation about 34x larger	MR evidence answers a relation question distinct from accuracy	The quantities are different relative-L2 objects
Symmetric mesh	Exact mirror-y admitted and fails; relative L2 1.10	The out-of-sample gate admits the relation when geometry is bijective	Synthetic no-obstacle OOD geometry
Seeded faults	5/10 mutants detected; detections separate by MR family	Admitted MRs expose some invariant-breaking faults and blind spots	Stress-test evidence, not a real-world defect-detection rate

and an OpenMC subject run. Fault coverage follows each program's admitted MR set; structural blind spots remain in six of seven programs; and class-to-class mappings are program-specific. This supports the falsification role of the admitted-set view. It is a read-only check from committed evidence, not an end-to-end execution of this pipeline on each program (claim C39).

End-to-end cross-program execution. The same gate and typed verdict are also executed on five CPU-only sibling SUTs: heat, wave, point-kinetics, Burgers, and the sibling's OpenMC single-energy-group subject run. Across the five, the gate admits 38 candidate relations and rejects 7 at output-mapping conditions; admitted relations hold on correct programs, while mutant runs populate the SUT-inconsistency region. Detection tracks the admitted set rather than a uniform rate (full-set mutation score 0.75 on heat, 0.20 on the other three solvers, and 0.56 on OpenMC). The MRs and mutants are the sibling's; only the gate and typed verdict are newly executed here, so this asserts no per-program reliability, no superiority, no new MR contribution, and no continuous-energy or whole-core reactor-physics result (claim C40).

Table A6: Interpretation and claim-boundary guide.

Reader concern	Evidence response	Licensed claim	Not licensed
Does MR evidence add value beyond accuracy?	Rollout accuracy and mirror-y violation measure different objects; seeded faults include MR-only catches	Relation-indexed checks complement accuracy monitoring	admitted relation set superiority over accuracy monitors
Is the gate merely post-hoc labeling?	Symmetric-mesh, airfoil, PINN, FNO, and witness reruns apply the same predicate outside the first pilot	The predicate changes or preserves verdicts according to relation conditions	General reliability from one MeshGraphNets family
Does downgrading hide robustness failures?	OOD verdicts remain reported as stress or robustness evidence, but not as valid-domain fault verdicts	Gate controls fault attribution	Using domain invalidity to excuse deployment weakness
Is deferred conservation a pipeline failure?	FNO periodic conservation and PINN reference-relative conservation produce executable verdicts where the floor is calibratable	The gate fails closed when numerical decidability is missing	Absolute MGN mass conservation on the coarse mesh
What overclaims are blocked?	Typed verdicts distinguish failure, OOD stress, rejection, deferral, and inconclusive evidence	Bounded V&V utility with auditable artifacts	Geometry-independent rates, validated localization, or real-world defect detection
When is the method worth using?	A relation must be physically meaningful and numerically decidable for the deployed discretization	Use for relation-indexed OOD or physics checks alongside accuracy/UQ	Replacement for residual, UQ, or accuracy diagnostics
What is the cost profile?	Relation-record authoring and floor analysis are one-time per-relation design costs; checks then replay automatically	Reusable executable checks with visible assumptions	Zero-cost automatic MR discovery or measured person-hour savings

4. Secondary baselines, external-scope audit, and cross-family executions

4.1. RQ4 secondary baselines and external-scope audit

This subsection reports the Secondary baseline and external-scope audit; LLM baselines are secondary exploratory scope contrasts, not competitors. Expert-LLMs proposed 24 unique candidates, admissibility voting retained 4 with no novel family, generic templates admitted 3/13, and one-shot rater agreement was Fleiss kappa 0.077. These numbers concern candidate screening, not the deterministic operator-floor admissibility condition. On the converged PointMLP SUT, the admitted relation set and rollout-accuracy monitor are complementary over a 20-fault catalogue (relation set 13/20, monitor 6/20; MR-only 9, accuracy-only 2, both 4, neither 5). The single gate-admitted generic template has 0% baseline false positives on the correct SUT, whereas all six gate-rejected templates fire on it (100%). This is a measured gate value with descriptive Wilson CIs and no superiority test. The Minimum-MR-SubSet external witness evidence records 70 real witness instances, three SciML/PDE rows with PASS_WITNESS status, and one held-out cylinder-flow MGN rerun; it does not add new primary SUT executions to this paper.

4.2. RQ4 Cross-family PINN extension and FNO upgrade

The cross-family executions have a supporting falsification role. They test whether the same four admissibility conditions produce typed decisions on pointwise PINNs and spectral FNOs, and whether conservation MRs execute wherever the floor is calibratable: FNO periodic conservation fails 24/24 against a case-level floor and PINN reference-relative conservation passes 30/30, while the MGN open-boundary absolute variant is refused. The cylinder-flow deferral is therefore gate discrimination rather than missing verdict evidence.

The K=15 PINN roster contains fifteen Burgers and fifteen heat seeds. **MR-A** is *vacuous by construction* for pointwise MLPs. The two non-trivial MRs carry the evidence: **MR-B** passes on 13/15 Burgers seeds (mean 0.712, CI [0.583, 0.875]) and is mixed on heat with 7/15 passing (mean 1.495, CI [1.142, 1.897]); **MR-C** passes on all 15/15 of both PDEs, with Burgers mean 1.007 (CI [0.997, 1.017]) and heat mean 1.006 (CI [0.988, 1.022]). This is a two-PDE PINN seed roster rather than a PINN-vs-MGN benchmark or a generality claim across PINN architectures, PDEs, or training regimes.

The **FNO primary workflow upgrade** uses six trained FNO-2D checkpoints over Burgers and heat. Periodic integer translation is admitted and produces **24/24 translation passes** (maximum relative-L2 violation below 10^{-5}). The **periodic discrete-conservation MR** is admitted-with-reference-floor, and trained FNO outputs exceed that calibrated floor on **24/24 conservation failures**. The Dirichlet translation candidate is rejected for 6/6 SUTs because it changes the boundary-value problem and is not executed as an exact MR. This is **not only admissibility evidence**: it is full gate-to-verdict FNO evidence with raw source/follow-up outputs and per-case ledgers, while remaining outside cylinder-flow evidence, performance benchmarking, reliability, and broad neural-operator generalization. The separate FNO admissibility roster is admitted on 15/15 per PDE (Wilson 95% CI $[0.80, 1.00]$), with Burgers mean violation 4.5×10^{-8} and heat 1.4×10^{-7}.

4.3. Production-adjacent RealPDEBench foil preflight

The supplemental RealPDEBench check is a preflight witness, not a new SUT pass/fail result. Metadata screening found machine-readable angle of attack and Reynolds number fields, zero-AoA real and numerical cases, and stored `x`/`y`/`u`/`v`/`t` fields for the public foil scenario. The first downloaded zero-AoA real-flow Arrow shard corresponds to `10000_0.0.h5`; its coordinate mirror map is exact on the stored grid (x-invariance floor 0.0 and y-reflection floor 0.0). Mirroring and sign-reversing the real velocity field gives u relative L2 0.0600, sign-reversed-v relative L2 1.3753, and combined uv relative L2 0.2292. Because this preflight has no independent PIV measurement-floor bound, the typed verdict is *inconclusive*: the evidence shows that a public real-flow benchmark can be brought through metadata screening, coordinate mapping, and metric generation, but it does not license a trained-SUT correctness, real-defect, production-CFD, or broad aerodynamic-generalization claim (claim C55).

4.4. External issue/PR/commit-linked witness corpus

This supplemental corpus links five public external issue/PR/commit records to full external witness checks. Each counted unit has archived source artifacts, a relation record, a source/follow-up or before/after semantic check, a metric, and a typed verdict. The aggregate report records 5/5 *pass* typed verdicts across four repositories or independent subsystems (claim C57). The corpus is designed to answer the self-made-task objection; it is not a representative sample of SciML defects, a trained-SUT correctness study, production validation, or a defect-rate estimate.

The following reviewer-facing screening protocol defines how the corpus should be read. It is reviewer-facing external evidence: it covers several software components outside the self-made tasks, but it is not statistically representative.

Reviewer-facing screening protocol.. **Inclusion criteria.** Counted units required a public external issue, PR, commit, or maintainer-linked source artifact; SciML or scientific-software relevance; a clear semantic relation representable as a relation record and checked by the gate; a source/follow-up or before/after semantic contrast; a CPU-feasible local witness; a concrete metric; a typed verdict; and claim-ledger/experiment-ledger traceability. **Exclusion criteria.** We excluded ordinary API/runtime failures without a clear semantic relation, hardware/tolerance-only instability without a deterministic contrast, user questions without fix evidence, dependency-heavy cases that could not be replayed in the one-week JSS risk-reduction scope, and non-SciML download or preprocessing issues. **Purposeful screen.** The screen was purposeful rather than random sampling from a complete defect population. The screening ledger therefore records go, defer, and no-go categories, but it does not license a defect-prevalence estimate.

Component coverage map.. The five counted witnesses cover different SciML semantic components: DeepXDE covers periodic boundary-condition derivative semantics; NeuralOperator `spectrum_2d` covers spectral metric numerical-decidability semantics; NeuralOperator SpectralConv covers Hermitian frequency-domain symmetry semantics; PhiFlow/PhiML covers coordinate/component axis-order gradient semantics; and JAX-CFD covers advection flux boundary-condition inference semantics. This breadth addresses the concern that the workflow only works on self-made tasks, but it remains a curated external witness corpus rather than a statistically representative real-defect corpus.

Evidence ladder and claim boundary.. The evidence ladder starts with author-designed or synthetic task evidence, then adds independent SUT/task full-chain evidence, and then adds an external issue/PR/commit-linked witness corpus across multiple repositories or subsystems. A stronger future tier would require a production-scale or statistically representative real-defect corpus under a defined sampling frame. The current paper reaches the external-witness tier, not that stronger future tier. It allows an external-witness claim; it forbids production validation, trained-SUT correctness, defect rate, defect prevalence, and representative sampling claims.

The DeepXDE PeriodicBC issue-linked witness remains the seed unit: PR #27 `derivative_order=1` residual is 2.0 on `u=x(1-x)`, while the pre-PR value-only residual is 0.0 (claim C56).

Table A7: External issue/PR/commit-linked witnesses.

Unit	**External source**	**Semantic metric**	**Boundary**
DeepXDE PeriodicBC	Issue #26 and merged PR #27	`u=x(1-x)` has value residual 0.0 but `derivative_order=1` residual 2.0; typed pass	Boundary-condition component witness only.
NeuralOperator `spectrum_2d`	Issue #532 and merged PR #661	Old L1 radius collides modes (2,0) and (1,1); corrected L2 gap is 0.5858; old power after complex sum is 0.0 versus corrected squared-magnitude sum 2.0; typed pass	Spectral utility witness, not trained-FNO reliability.
NeuralOperator SpectralConv	Merged PR #702	Hermitian `irfft` boundary-frequency imaginary parts drop from max 0.5 to 0.0 after DC/Nyquist enforcement; typed pass	Does not reproduce GPU line artifacts locally.
PhiFlow/PhiML custom gradient	PhiFlow issue #199 and PhiML commit `96ef3e4d...`	Old native gradient order remains shape [2,3]; name-aware corrected native order is [3,2] and matches expected axis order; typed pass	Custom-gradient axis-order witness, not full flow simulation validation.
JAX-CFD advection flux	Merged PR #167	Old scalar-BC inheritance gives Neumann flux at a no-penetration wall; corrected velocity/scalar inference gives Dirichlet flux; typed pass	Boundary-condition inference witness, not solver correctness.